Phase-I Review-II / Presentation-2 Report

Reliable Review Allocation Under Distribution Shift

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Chapter 1

Abstract

This project presents a reliable review allocation framework for machine learning systems operating under distribution shift. In practical machine learning deployments, prediction confidence alone may not reliably indicate when a model is likely to make an error because the distribution of incoming data can differ from the training distribution.

The proposed framework combines multiple risk signals, including prediction confidence, out-of-distribution (OOD) indicators, distribution-drift information, and historical error patterns. These signals are combined into an adaptive risk score that prioritizes instances for human review under a fixed review budget.

The system is evaluated using a network intrusion detection setting with a chronological train-validation-test split. Several review allocation strategies are compared, including random selection, confidence-based selection, OOD-based selection, static multi-factor allocation, and adaptive multi-factor allocation. Evaluation is performed at review budgets of 1%, 2%, 5%, and 10%.

The results show that adaptive allocation provides better error capture than confidence-based allocation at the smaller review budgets, while confidence-based allocation performs better at wider budgets in the current experiment. This demonstrates that the effectiveness of a review strategy depends on the available human-review capacity and the characteristics of the distribution shift.

The project focuses on reliable and leakage-controlled evaluation of human-review allocation rather than replacing the machine learning model itself. Ground-truth labels are not used during ranking; they are used only after the review queue has been selected for evaluation and simulated feedback. The resulting framework provides a reproducible basis for studying selective human review under changing data distributions.

Chapter 2

Introduction

2.1 Introduction

The deployment of machine learning algorithms in operational systems has expanded from advisory tools to automated decision infrastructure. Across enterprise cybersecurity, payment networks, and healthcare diagnostics, automated decision systems process large streams of transactions and network events daily. However, the foundational statistical assumption governing supervised machine learning—that training and operational test data are independent and identically distributed (i.i.d.)—is routinely challenged in real-world deployments. Network traffic patterns evolve, threat actors adapt evasion strategies, and operational infrastructure changes over time, introducing substantial covariate and concept drift.

Chapter 3

Literature Review

3.1 Detailed Literature Survey

A rigorous literature survey was conducted across 10 foundational research papers related to distribution shift, selective prediction, learning to defer, uncertainty estimation, human-in-the-loop machine learning, and network intrusion detection. Each paper was examined to understand its objective, methodology, experimental setting, limitations, and relevance to the proposed study.

3.2 Research Paper 1

Anomaly Detection Under Distribution Shift

Authors: T. Cao, J. Zhu, and G. Pang

Year: 2023

Venue: IEEE/CVF International Conference on Computer Vision (ICCV 2023), pp. 6511–6523.

DOI: `https://doi.org/10.1109/ICCV51070.2023.00599`

This work addresses the degradation of anomaly detection models when operational test environments exhibit covariate and semantic distribution shift relative to training data. The study focuses on the problem of distinguishing genuine anomalies from variations caused by changes in the data distribution.

The proposed approach combines dual-memory contrastive feature learning with distribution discrepancy minimization. The method is evaluated on multiple visual benchmark datasets exhibiting different forms of distribution shift.

The study establishes the importance of evaluating anomaly detection methods under distribution shift. However, it primarily focuses on high-dimensional computer-vision representations and does not address capacity-constrained human-review queues for tabular network-security data.

The work is relevant to the present project because it supports the fundamental motivation that distribution shift can reduce the reliability of machine-learning predictions.

3.3 Research Paper 2

Consistent Estimators for Learning to Defer to an Expert

Authors: H. Mozannar and D. Sontag

Year: 2020

Venue: International Conference on Machine Learning (ICML 2020), PMLR vol. 119, pp. 7076–7087.

This work investigates human-AI collaborative decision-making and the problem of learning when a machine should defer a prediction to a human expert.

The authors formulate learning-to-defer as an extended classification problem with a rejection option and develop a consistent surrogate loss for jointly learning classification and deferral decisions.

The study is evaluated using synthetic mixture models and medical diagnosis datasets with simulated expert profiles.

The work provides an important theoretical foundation for human-AI deferral. However, it primarily assumes stationary data distributions and does not directly address strict hard review-capacity quotas such as 1%, 2%, 5%, or 10%.

The present project extends the practical interpretation of deferral toward a ranking-based review-allocation problem under temporal distribution shift and fixed review capacity.

3.4 Research Paper 3

Exploiting Human-AI Dependence for Learning to Defer

Authors: Z. Wei, Y. Cao, and L. Feng

Year: 2024

Venue: International Conference on Machine Learning (ICML 2024), PMLR vol. 235, pp. 52484–52499.

This study examines the dependence between machine predictions and human expert decisions in learning-to-defer systems.

The authors develop a deferral architecture that explicitly models the relationship between human and machine decisions rather than assuming that expert performance is independent of machine errors.

The approach is evaluated using image-classification and text-detection benchmarks involving simulated and real human annotation settings.

The study demonstrates that the relationship between machine errors and human decisions is important when designing reliable deferral systems. However, the approach requires human-machine labels during training and is therefore different from the post-hoc review-allocation setting considered in this project.

The work motivates the importance of using information about previous review outcomes when future feedback becomes available.

3.5 Research Paper 4

Selective Classification for Deep Neural Networks

Authors: Y. Geifman and R. El-Yaniv

Year: 2017

Venue: Advances in Neural Information Processing Systems (NeurIPS 2017), pp. 4878–4887.

This work formalizes selective classification, where a machine-learning system is allowed to reject or defer uncertain predictions rather than automatically making every decision.

The authors introduce SelectiveNet and investigate how model risk can be controlled at different coverage levels.

The method is evaluated on image-classification datasets including SVHN, CIFAR-10, and ImageNet.

The study demonstrates that selective prediction can reduce error risk when automated coverage is reduced. However, the experiments primarily consider standard i.i.d. settings rather than temporal distribution shift.

This work provides the conceptual basis for treating human review as a selective decision mechanism and motivates the use of confidence-based allocation as a baseline.

3.6 Research Paper 5

On Calibration of Modern Neural Networks

Authors: C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger

Year: 2017

Venue: International Conference on Machine Learning (ICML 2017), PMLR vol. 70, pp. 1321–1330.

This study investigates the calibration of modern neural networks and demonstrates that high classification accuracy does not necessarily imply reliable predicted probabilities.

The authors evaluate several post-hoc calibration techniques, including histogram binning, isotonic regression, and temperature scaling.

The study demonstrates that temperature scaling can substantially improve calibration without reducing classification accuracy on several benchmark datasets.

The work is relevant because prediction confidence is an important review-allocation signal in the present project. However, calibration behaviour can change when the operational distribution shifts, motivating the use of additional reliability signals beyond confidence alone.

3.7 Research Paper 6

Human-in-the-Loop Cyber Intrusion Detection Using Active Learning

Authors: Y. Kim, G. Dan, and Q. Zhu

Year: 2024

Venue: IEEE Transactions on Information Forensics and Security, vol. 19, pp. 8658–8672, 2024.

DOI: `https://doi.org/10.1109/TIFS.2024.3434647`

This work investigates human involvement in cyber-intrusion detection, where large alert volumes can exceed the available capacity of security analysts.

The study formulates analyst-query selection as an active-learning problem and considers uncertainty and data characteristics when selecting alerts for human inspection.

The work is particularly relevant to the operational domain of network intrusion detection because it demonstrates the importance of prioritizing security events for limited human attention.

However, active learning primarily focuses on acquiring labels for model improvement, whereas the present project focuses on allocating a fixed review budget for evaluating and prioritizing predictions from an existing classifier.

3.8 Research Paper 7

Incorporating Uncertainty in Learning to Defer for Safe Computer-Aided Diagnosis

Authors: J. Liu, B. Gallego, and S. Barbieri

Year: 2022

Venue: Scientific Reports, vol. 12, Art. no. 1762, 2022.

DOI: `https://doi.org/10.1038/s41598-022-05764-1`

This study investigates uncertainty-aware learning-to-defer systems for medical decision-making.

The approach incorporates uncertainty information into the deferral process so that potentially unreliable automated decisions can be redirected to human experts.

The study demonstrates that uncertainty can improve the safety of automated decision systems in selected medical tasks.

The work is relevant to the present project because it supports the use of multiple reliability indicators when deciding which predictions should receive human attention.

However, the computational requirements and domain-specific assumptions differ from the lightweight tabular network-flow setting considered in this project.

3.9 Research Paper 8

Generalized Out-of-Distribution Detection: A Survey

Authors: J. Yang, K. Zhou, Y. Li, and Z. Liu

Year: 2024

Venue: International Journal of Computer Vision, vol. 132, no. 12, pp. 5635–5662, 2024.

DOI: `https://doi.org/10.1007/s11263-024-02117-4`

This survey provides a broad review of out-of-distribution detection methods and discusses different approaches for identifying observations that differ from the training distribution.

The work categorizes OOD techniques into output-based, feature-based, and density-based approaches and discusses their application across multiple domains.

Mahalanobis distance is identified as an important feature-space approach for measuring the distance between an observation and a reference distribution.

The present project uses a regularized Mahalanobis-distance-based signal as one component of its reliability framework. The survey also highlights an important distinction between detecting novelty and detecting prediction errors.

3.10 Research Paper 9

WILDS: A Benchmark of In-the-Wild Distribution Shifts

Authors: P. W. Koh et al.

Year: 2021

Venue: International Conference on Machine Learning (ICML 2021), PMLR vol. 139, pp. 5637–5664.

This work introduces WILDS, a benchmark designed to evaluate machine-learning models under naturally occurring distribution shifts.

The benchmark contains multiple datasets with real-world shifts involving different domains, locations, time periods, and populations.

The study demonstrates that models trained using conventional empirical-risk-minimization approaches can experience substantial performance degradation under distribution shift.

The work directly motivates the use of chronological evaluation in the present project. Rather than relying only on a random i.i.d. split, the project uses a later chronological partition to evaluate review allocation under a changed operational distribution.

3.11 Research Paper 10

Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey

Authors: K. He, D. D. Kim, and M. R. Asghar

Year: 2023

Venue: IEEE Communications Surveys & Tutorials, vol. 25, no. 1, pp. 538–566, 2023.

DOI: `https://doi.org/10.1109/COMST.2022.3233793`

This survey examines vulnerabilities of machine-learning-based network intrusion detection systems under adversarial attacks and changing operational conditions.

The study discusses adversarial evasion, poisoning, and other threats that can cause intrusion-detection models to produce incorrect predictions.

The work is relevant to the present project because changes in network behaviour and adversarial strategies can contribute to situations where historical model reliability no longer represents future operational behaviour.

However, the survey does not directly formulate a capacity-constrained human-review allocation mechanism. The present project addresses this complementary problem by focusing on how limited review capacity can be allocated among model predictions.

3.12 Literature Gap and Research Motivation

The reviewed studies establish several important foundations.

First, distribution shift can substantially reduce the reliability of machine-learning models when deployment data differ from training data. Second, selective prediction and learning-to-defer demonstrate that uncertain predictions can be redirected for additional human attention. Third, uncertainty and OOD methods provide different forms of reliability information. Finally, research in network intrusion detection demonstrates the practical importance of managing large volumes of security alerts.

However, the reviewed literature provides limited direct investigation of the combined problem considered in this project: allocating a fixed and explicitly constrained human-review budget using multiple reliability signals under temporal distribution shift.

The present study therefore evaluates Random, Confidence-only, OOD-only, Static Multi-Factor, and Adaptive Multi-Factor review-allocation strategies under identical review budgets. The objective is not to claim universal superiority of the adaptive approach, but to experimentally determine whether combining multiple reliability signals improves error concentration when review capacity is highly constrained.

Chapter 4

Background

4.1 Machine Learning Under Distribution Shift

Machine-learning systems are commonly developed under the assumption that the data observed during deployment are sufficiently similar to the data used during training. In operational environments, this assumption can be violated because data distributions change over time.

For network intrusion detection, changes may occur because of new applications, changing network behaviour, infrastructure changes, or changes in attack patterns. A model that performs well on historical data can therefore experience increased prediction errors when applied to a later distribution.

The present project focuses specifically on this temporal distribution-shift setting and evaluates whether reliability information can be used to prioritize predictions for limited human review.

4.2 Human Review and Selective Prediction

Human review can provide an additional decision layer for machine-learning systems. Instead of requiring a human to inspect every prediction, a system can identify a subset of predictions that are considered more uncertain or risky.

This idea is closely related to selective classification and learning-to-defer. In a selective prediction setting, the system attempts to maintain reliable automated decisions while identifying cases that should not be handled automatically.

The challenge considered in this project is different from simply selecting a rejection threshold. The available review capacity is explicitly constrained to fixed percentages of the prediction stream.

4.3 Review Capacity as a Resource Constraint

Human review requires analyst time and therefore cannot necessarily be applied to every prediction in a high-volume environment.

Let N represent the number of predictions in a batch and B represent the available review percentage. The maximum review capacity is:

$$K = \lfloor B \times N \rfloor \tag{4.1}$$

The system must therefore decide which K predictions should receive review.

This transforms the problem into a ranking problem: the objective is to place predictions that are more likely to contain model errors near the top of the review queue.

4.4 Prediction Confidence

Prediction confidence is one of the simplest signals that can be used for review prioritization.

For a binary classifier, predictions close to the decision boundary generally have lower confidence than predictions with probabilities close to zero or one. The confidence-risk score used in this project is:

$$s_{\text{conf}} = 1 - \max(P(y = 0|x), P(y = 1|x)) \tag{4.2}$$

A higher value represents greater uncertainty and therefore higher review priority.

Confidence-based allocation is used as an important baseline in the project.

4.5 Out-of-Distribution Detection

Out-of-distribution detection attempts to identify observations that differ from the distribution represented by the training data.

The project uses a regularized Mahalanobis distance to measure the geometric distance of an observation from the training feature distribution:

$$d_M(x) = \sqrt{(x-\mu)^T(\Sigma+\lambda I)^{-1}(x-\mu)} \tag{4.3}$$

where μ represents the training-data mean vector, Σ represents the covariance matrix, and λI provides regularization.

A high OOD score indicates greater feature-space novelty. However, novelty does not necessarily mean that the classifier prediction is incorrect. This distinction is important when interpreting the OOD-only allocation results.

4.6 Distribution Drift

Distribution drift refers to changes in the statistical characteristics of data over time.

The project uses Population Stability Index (PSI) to measure feature-level distribution changes between a reference distribution and a later distribution.

The resulting drift information is then incorporated into an instance-level drift-risk signal. This allows the allocation mechanism to consider not only whether an individual prediction is uncertain, but also whether the surrounding data distribution has changed.

4.7 Historical Model Errors

A further reliability signal is obtained from previous model mistakes.

The implementation identifies prediction errors made on the training data and maps these errors to feature-based regions. New observations that resemble regions associated with previous mistakes receive a higher historical-error risk.

This signal is intended to complement confidence, OOD, and drift information by capturing regions in the feature space where the model has previously demonstrated weaker performance.

4.8 Multi-Factor Reliability Assessment

The central idea of the project is to combine multiple reliability signals instead of relying on a single indicator.

The composite risk score is represented as:

$$R_i = w_c s_{\text{conf},i} + w_o s_{\text{ood},i} + w_d s_{\text{drift},i} + w_h s_{\text{hist},i} \tag{4.4}$$

where the four terms represent confidence, OOD, drift, and historical-error risk respectively.

The resulting score is used to rank predictions for review.

4.9 Relevance to the Proposed Study

The background concepts provide the foundation for the experimental comparison performed in this project.

Confidence-only allocation represents a single-signal strategy. OOD-only allocation evaluates feature-space novelty independently. Static multi-factor allocation combines multiple signals using fixed weights, while adaptive multi-factor allocation changes the weighting according to the observed drift regime.

The project therefore evaluates whether combining reliability information provides an advantage when review capacity is particularly limited.

Chapter 5

Comparative Study

5.1 Overview

The project compares multiple review-allocation strategies under the same chronological shifted-test setting and the same review-capacity constraints.

The comparison is designed to determine whether combining multiple reliability signals provides an advantage over simpler single-signal approaches.

5.2 Allocation Strategies

Five review-allocation strategies are evaluated.

5.2.1 Random Allocation

Random allocation selects review instances without using the reliability signals.

It provides a reference baseline representing the expected performance when review capacity is distributed without risk-based prioritization.

5.2.2 Confidence-Only Allocation

Confidence-only allocation ranks instances according to prediction uncertainty.

The confidence-risk score is defined as:

$$s_{\text{conf}} = 1 - \max(P(y = 0|x), P(y = 1|x)) \tag{5.1}$$

Instances with higher confidence risk are prioritized for review.

This strategy represents a conventional approach in which the model's own prediction uncertainty is the primary indicator of review priority.

5.2.3 OOD-Only Allocation

OOD-only allocation ranks instances according to their distance from the training feature distribution.

The project uses a regularized Mahalanobis distance and converts the resulting values into percentile-based OOD scores.

This strategy tests whether feature-space novelty alone is sufficient for identifying erroneous predictions under distribution shift.

5.2.4 Static Multi-Factor Allocation

Static multi-factor allocation combines the four reliability signals using fixed weights.

The score is represented as:

$$R_{\text{static}} = w_c s_{\text{conf}} + w_o s_{\text{ood}} + w_d s_{\text{drift}} + w_h s_{\text{hist}} \tag{5.2}$$

The same weighting scheme is maintained across the evaluated batches.

5.2.5 Adaptive Multi-Factor Allocation

Adaptive multi-factor allocation also combines the four reliability signals, but the weights are changed according to the observed drift regime.

The implementation uses transparent heuristic weight configurations for low-, moderate-, and high-drift regimes.

The adaptive strategy is therefore intended to test whether changing the relative importance of the reliability signals according to the observed data regime can improve review prioritization.

5.3 Comparison Criteria

All five strategies are evaluated using the same shifted-test data and the same review budgets.

The evaluated budgets are:

$$B \in \{1\%, 2\%, 5\%, 10\%\} \tag{5.3}$$

For every strategy, the review queue contains the same maximum number of instances for a given budget.

This makes the comparison a constrained allocation problem rather than a comparison based on different amounts of human-review capacity.

5.4 Evaluation Measures

The primary comparison measure is error capture rate.

$$ECR = \frac{\text{Errors Captured}}{\text{Total Model Errors}} \times 100 \tag{5.4}$$

Review efficiency is also measured using errors captured per 1,000 reviewed instances:

$$Efficiency = \frac{\text{Errors Captured}}{\text{Reviewed Instances}} \times 1000 \tag{5.5}$$

These measures evaluate both the effectiveness of the selected review queue and the efficiency with which the limited review capacity is used.

5.5 Comparison of Adaptive and Confidence Strategies

The current shifted-test results show the following error-capture values:

Table 5.1: Comparison of Adaptive and Confidence-Based Allocation

Review Budget	Adaptive (%)	Confidence (%)
1%	17.80	13.51
2%	34.26	26.73
5%	45.90	65.45
10%	52.35	84.01

The adaptive strategy performs better than confidence-only allocation at the 1% and 2% budgets in the current experiment.

At larger review budgets, confidence-based allocation performs better. At 5%, confidence captures 65.45% of model errors compared with 45.90% for the adaptive strategy. At 10%, confidence captures 84.01% compared with 52.35% for adaptive allocation.

The comparison therefore indicates that the usefulness of the multi-factor adaptive mechanism is dependent on the available review capacity.

5.6 Interpretation of the Single-Signal Strategies

The confidence-only and OOD-only strategies provide useful comparisons because they isolate individual sources of reliability information.

Confidence directly reflects uncertainty in the classifier's output, while OOD measures feature-space novelty.

These signals represent different concepts. A prediction can be highly confident while still being incorrect, particularly when the operational distribution differs from the training distribution. Conversely, an observation can be highly novel while still receiving a correct prediction.

The comparison therefore motivates the investigation of multiple complementary signals rather than assuming that any single signal is universally sufficient.

5.7 Comparison Under Distribution Shift

The evaluation is performed using a chronological shifted-test partition rather than a randomly sampled i.i.d. test set.

This allows the allocation strategies to be evaluated under a later data distribution and provides a more challenging setting for confidence-based reliability estimation.

The results demonstrate that distribution shift can change the relative effectiveness of different review-prioritization signals.

5.8 Summary of Comparative Findings

The comparative study produces the following observations:

1. Random allocation provides a non-risk-based reference baseline.
2. Confidence-only allocation provides a strong baseline, particularly at larger review budgets in the current experiment.
3. OOD-only allocation tests feature-space novelty independently but does not imply prediction error directly.
4. Static multi-factor allocation evaluates whether combining reliability signals is useful without changing their weights.
5. Adaptive multi-factor allocation provides its strongest observed advantage at the 1% and 2% review budgets.
6. No single strategy is demonstrated to dominate across all evaluated review budgets.

The comparative results therefore support a budget-dependent interpretation of review allocation under distribution shift.

Chapter 6

Requirements

6.1 Overview

The requirements define the functional behaviour and quality expectations of the proposed review-allocation system. The requirements are limited to the approved research scope of evaluating constrained human-review allocation under distribution shift.

6.2 Functional Requirements

The system shall satisfy the following functional requirements:

1. **Dataset Ingestion:** The system shall load the selected network-flow dataset and prepare the data for experimentation.
2. **Chronological Partitioning:** The system shall preserve temporal ordering and divide the experimental data into training, validation, and shifted-test partitions.
3. **Data Preprocessing:** The system shall handle invalid and missing numerical values using preprocessing parameters derived from the training partition.
4. **Model Training:** The system shall train and evaluate Logistic Regression, Random Forest, and XGBoost classification models.
5. **Model Selection:** The system shall select the primary model using validation performance.
6. **Prediction Generation:** The system shall generate binary predictions and class probabilities for the evaluation data.
7. **Confidence Signal Generation:** The system shall calculate prediction-confidence risk from model probabilities.
8. **OOD Signal Generation:** The system shall calculate a regularized Mahalanobis-distance-based OOD signal.
9. **Drift Signal Generation:** The system shall calculate feature-level distribution drift using PSI and derive an instance-level drift-risk score.
10. **Historical-Error Signal Generation:** The system shall identify training-set prediction mistakes and use feature-based error patterns to construct a historical-error risk signal.
11. **Risk-Score Generation:** The system shall combine the reliability signals into a composite risk score.
12. **Review Allocation:** The system shall rank predictions according to risk and allocate only the permitted number of predictions to the review queue.
13. **Strategy Comparison:** The system shall compare Random, Confidence-only, OOD-only, Static Multi-Factor, and Adaptive Multi-Factor allocation strategies.
14. **Evaluation and Visualization:** The system shall calculate review-allocation results and provide a local exploratory dashboard for examining the experimental outputs.

6.3 Non-Functional Requirements

6.3.1 Reproducibility

The experimental pipeline should produce reproducible results when executed with the same data, configuration, and random-state settings.

6.3.2 Data Leakage Control

Information from the shifted-test labels shall not be used to construct the initial review ranking. Ground-truth information is used for offline evaluation after the review queue has been selected.

6.3.3 Modularity

The implementation should maintain separate processing stages for preprocessing, model training, reliability-signal generation, risk scoring, allocation, and evaluation.

6.3.4 Interpretability

The reliability signals and risk-score formulation should remain sufficiently transparent to allow their contribution to the review-allocation decision to be examined.

6.3.5 Capacity Enforcement

Every allocation strategy shall operate under the same predefined review budget. The system shall not exceed the specified review capacity.

6.3.6 Scalability

The implementation should support processing large batches of network-flow records using vectorized numerical and data-processing operations where applicable.

6.3.7 Maintainability

The implementation should use separate scripts and structured result files so that individual experimental stages can be modified or evaluated independently.

6.3.8 Demonstrability

The project should provide executable evidence of the implemented pipeline through the Python modules, generated experiment outputs, and local Streamlit dashboard.

6.4 Hardware and Software Requirements

6.4.1 Software Requirements

The implementation requires:

- Python.
- Pandas and NumPy.
- Scikit-learn.
- XGBoost.
- Matplotlib.
- Joblib.
- Streamlit.
- A LaTeX environment for report generation.

6.4.2 Hardware Requirements

The experiments require a computer capable of processing large numerical datasets and executing Python-based machine-learning workloads.

Adequate system memory and storage are required because the project works with large network-flow datasets and intermediate result files.

A GPU is not a mandatory requirement for the implemented classical machine-learning pipeline.

6.5 Scope Boundary

The following capabilities are outside the current Phase-I implementation:

- Live network-traffic ingestion.
- Production deployment as a network intrusion detection service.
- Integration with a real human-review workforce.
- Automatic learning of adaptive weights from a large historical review database.
- Enterprise authentication and access-control infrastructure.
- Distributed production-scale serving.

These items may be considered future extensions but are not required to demonstrate the current research objective.

6.6 Implementation Status

The current implementation provides the major components required to demonstrate the research pipeline, including data preparation, model training, reliability-signal generation, review allocation, comparative evaluation, and local result visualization.

The feedback mechanism is treated as an experimental extension and must be demonstrated with a clear separation between ranking, label revelation, and subsequent feedback processing.

The implementation status should therefore be reported according to demonstrable completed modules rather than using an unsupported percentage claim.

Chapter 7

Methodology

7.1 Research Methodology

The proposed study evaluates human-review allocation for machine learning predictions under temporal distribution shift. The methodology follows a chronological experimental design in which historical data are used for model development and reliability-signal construction, while a later shifted partition is reserved for evaluating review-allocation strategies.

The overall workflow consists of four stages:

1. Chronological data partitioning, preprocessing, and baseline model training.
2. Extraction of confidence, OOD, drift, and historical-error signals.
3. Combination of the signals into a composite risk score and allocation of a fixed review budget.
4. Offline comparison of review-allocation strategies using a simulated ground-truth review oracle.

7.2 Dataset and Experimental Partitioning

The experiments use a 500,000-flow chronological sample from the BigFlow-NIDS benchmark. The benchmark contains approximately 66.9 million network-flow records and 55 attributes. The experimental data retain 49 numerical flow features and formulate the task as binary classification, where 0 represents benign traffic and 1 represents attack traffic.

The records are ordered chronologically using the flow timestamp and divided into three sequential partitions:

- Training set: 350,000 flows (70%).
- Validation set: 75,000 flows (15%).
- Shifted test set: 75,000 flows (15%).

The chronological split preserves the temporal structure of the data and allows the later test partition to represent a distribution that differs from the historical training period.

7.3 Data Preprocessing

Preprocessing is performed without using information from future partitions. Infinite values generated during flow-rate calculations are converted to missing values, and numerical missing values are imputed using medians calculated from the training set.

For Logistic Regression, StandardScaler is fitted exclusively on the training data and subsequently applied to the validation and shifted-test partitions. Tree-based models, including Random Forest and XGBoost, operate directly on the preprocessed numerical features.

This procedure prevents preprocessing parameters from the validation or shifted-test partitions from influencing model fitting.

7.4 Baseline Machine Learning Model

Three classification models are evaluated:

- Logistic Regression
- Random Forest
- XGBoost

The models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. XGBoost is selected as the primary model based on its validation ROC-AUC of 0.9228.

The selected classifier produces class probabilities for each instance. These probabilities are used to construct the confidence-risk signal and contribute to the subsequent multi-factor risk calculation.

7.5 Reliability Signal Extraction

Four complementary signals are constructed to estimate the review priority of individual predictions.

7.5.1 Prediction Confidence

For a binary prediction, confidence-based risk is calculated as:

$$s_{\text{conf},i} = 1 - \max(P_i, 1 - P_i) \tag{7.1}$$

where P_i is the predicted probability of the attack class for instance i. Higher values indicate predictions closer to the decision boundary and therefore greater uncertainty.

7.5.2 Mahalanobis Out-of-Distribution Signal

A regularized Mahalanobis distance measures the geometric distance of an observation from the training feature distribution:

$$d_M(x_i) = \sqrt{(x_i - \mu)^T (\Sigma + \lambda I)^{-1} (x_i - \mu)} \tag{7.2}$$

where x_i is the feature vector, μ is the training-data mean vector, Σ is the training covariance matrix, I is the identity matrix, and λ is a regularization parameter.

The resulting distances are converted into percentile-based OOD scores, with larger values representing greater feature-space novelty.

7.5.3 Population Stability Index and Drift Risk

Population Stability Index (PSI) is calculated across the numerical feature space to identify distributional changes between reference and later data. Ten bins are used for the feature-level PSI calculation.

The resulting drift information is converted into instance-level risk using standardized feature deviations. Features exhibiting stronger distributional change contribute more strongly to the resulting drift-risk score.

7.5.4 Historical Error Risk

Historical-error risk is constructed from model prediction mistakes observed on the training data. The trained model generates predictions for the training partition, and instances where predictions differ from their training labels are identified as historical errors.

Training features are divided into feature-based regions using training-data quantiles. Historical errors are mapped to these regions, allowing new observations resembling regions with previous model mistakes to receive higher historical-error risk.

The resulting score is normalized before combination with the other reliability signals.

7.6 Composite Risk Score

The four reliability signals are combined into a composite review-priority score:

$$R_i = w_1 s_{\text{conf},i} + w_2 s_{\text{ood},i} + w_3 s_{\text{drift},i} + w_4 s_{\text{hist},i} \tag{7.3}$$

where the weights satisfy:

$$\sum_{j=1}^{4} w_j = 1 \tag{7.4}$$

The composite score provides a unified ranking of instances according to their estimated review priority.

The implementation uses transparent heuristic weights for different observed drift regimes. These weights are treated as allocation parameters rather than learned machine-learning model parameters.

7.7 Human-Review Allocation

After calculating the composite risk score, instances are ranked in descending order of risk. A fixed review budget is then enforced using:

$$B \in \{1\%, 2\%, 5\%, 10\%\} \tag{7.5}$$

For a batch containing N instances, the review capacity is:

$$K = \lfloor B \times N \rfloor \tag{7.6}$$

Only the highest-risk K instances are selected for review. The remaining instances remain in the autonomous prediction queue. This hard quota ensures that all strategies are compared under the same review capacity.

7.8 Review-Allocation Strategies

The proposed adaptive mechanism is compared with four baseline strategies:

1. Random allocation.
2. Confidence-only allocation.
3. OOD-only allocation.
4. Static multi-factor allocation.
5. Adaptive multi-factor allocation.

Random allocation provides a reference baseline, while confidence-only and OOD-only strategies test individual reliability signals. Static multi-factor allocation combines multiple signals using fixed weights, whereas adaptive multi-factor allocation changes the weighting according to the observed drift regime.

7.9 Simulated Review Oracle

The Phase-I implementation does not use live human reviewers. Instead, a simulated ground-truth review oracle is used for controlled offline benchmarking.

The review queue is first created using only the available reliability signals. After the queue has been finalized, ground-truth labels are used to determine whether the selected predictions are

erroneous. This allows the number of errors captured by each strategy to be measured under identical review budgets.

The ground-truth labels therefore do not form part of the initial risk ranking.

7.10 Evaluation Metrics

The primary evaluation measures how effectively each strategy identifies prediction errors within the available review budget.

The error capture rate is:

$$ECR = \frac{\text{Errors Captured}}{\text{Total Errors}} \times 100 \tag{7.7}$$

Review efficiency is measured as the number of errors captured per 1,000 reviewed instances:

$$Efficiency = \frac{\text{Errors Captured}}{K} \times 1000 \tag{7.8}$$

These measures are calculated independently for each allocation strategy and review budget.

7.11 Reproducibility and Leakage Control

The experimental pipeline uses deterministic random-state configuration and chronological data ordering to support reproducibility.

Preprocessing parameters are fitted only on the training partition. Model selection is performed using the validation partition, while the shifted-test partition is reserved for final offline evaluation.

The historical-error signal is constructed exclusively from training-set prediction mistakes. Validation and shifted-test labels are not used to construct this signal.

Similarly, the initial review ranking does not use shifted-test labels. Ground-truth information becomes available only after the review queue has been selected, allowing the selected cases to be evaluated by the simulated review oracle.

No model selection or weight optimization is performed using shifted-test labels.

7.12 Workflow Boundary

The current Phase-I methodology separates the experimental workflow into two implemented stages and one future operational stage.

7.12.1 Offline Benchmarking

Historical data are chronologically partitioned into training, validation, and shifted-test sets. Models are trained using the training partition, reliability signals are generated, risk scores are calculated, and review queues are constructed under fixed capacity constraints.

The shifted-test labels are not used during initial signal generation or ranking. They are used only after queue selection for offline evaluation of error capture.

7.12.2 Simulated Reviewer Feedback

After a review queue has been finalized, the selected top-K instances are evaluated using their ground-truth labels. The resulting information is used to calculate error concentration and review-allocation performance.

This simulated reviewer represents a controlled experimental oracle and should not be interpreted as evidence from actual human reviewers.

7.12.3 Future Operational Workflow

A live operational reviewer workflow is outside the scope of the current Phase-I implementation. A future extension could involve streaming predictions, continuous reliability-signal computation, real-time queue management, actual human reviewers, and incorporation of reviewer feedback into adaptive allocation.

The current results are therefore interpreted as evidence from controlled offline experiments rather than from a live operational environment.

7.13 PSI-to-Instance-Level Drift Conversion

PSI is a feature-level measure of distribution change, while review allocation requires an instance-level priority score. The implementation therefore converts batch-level drift information into instance-level risk.

Features with PSI values meeting the selected drift threshold are treated as drift-relevant. For each observation, standardized deviations from training statistics are calculated for these features. The deviations are weighted using the corresponding feature-level PSI values and aggregated to obtain an instance-level drift-risk score.

This conversion uses reference statistics and drift information rather than shifted-test ground-truth labels.

7.14 Algorithmic Execution

The complete experimental procedure is summarized below:

1. Load and chronologically order the BigFlow-NIDS sample.
2. Divide the data into training, validation, and shifted-test partitions.
3. Fit preprocessing parameters using the training partition.
4. Train and evaluate Logistic Regression, Random Forest, and XGBoost.
5. Select XGBoost using validation performance.
6. Generate prediction-confidence scores.
7. Calculate regularized Mahalanobis OOD scores.
8. Calculate PSI-based distribution-drift information.
9. Generate historical-error risk scores from training-set prediction mistakes.
10. Combine the reliability signals into the composite risk score.
11. Rank instances according to review priority.
12. Enforce the selected review quota.
13. Evaluate the selected queue using the simulated review oracle.
14. Compare all allocation strategies using error capture and review efficiency.

The resulting methodology provides a controlled framework for comparing human-review allocation strategies under temporal distribution shift while maintaining a fixed review capacity.

Chapter 8

Data Design

8.1 Dataset Overview

The project uses the BigFlow-NIDS benchmark for evaluating reliable review allocation under distribution shift. The complete benchmark contains approximately 66.9 million network-flow records with 55 attributes and 32 fine-grained attack categories.

For this study, a chronological sample of 500,000 network-flow records is used for experimental modelling. The selected feature space contains 49 numerical flow features. The detailed multi-class attack field is excluded from the modelling features so that the experimental task is formulated as binary classification.

8.2 Data Partitioning

The 500,000-flow experimental sample is divided chronologically into three partitions:

Table 8.1: Chronological Dataset Partition

Partition	Number of Flows	Percentage
Training	350,000	70%
Validation	75,000	15%
Shifted Test	75,000	15%
Total	500,000	100%

The training partition represents the earliest portion of the selected chronological sample. The validation partition follows the training period, while the shifted-test partition represents the later period used to evaluate performance under temporal distribution shift.

8.3 Feature Representation

The modelling data contain numerical network-flow characteristics such as duration, packet statistics, inter-arrival measurements, throughput-related measurements, TCP flags, and window-related attributes.

The 49 retained numerical features are used for model training and for constructing the reliability signals.

8.4 Target Representation

The prediction task is formulated as binary classification:

$$y_i = \begin{cases} 0, & \text{Benign traffic} \\ 1, & \text{Attack traffic} \end{cases} \tag{8.1}$$

The target is separated from the input feature matrix before model training.

8.5 Data Preprocessing Design

The preprocessing pipeline handles invalid and missing numerical values before model training.

Infinite values produced by flow-rate calculations are converted to missing values. Missing numerical values are then replaced using feature medians calculated from the training partition.

For Logistic Regression, feature scaling is performed using a StandardScaler fitted only on the training data. The same fitted transformation is subsequently applied to validation and shifted-test data.

Random Forest and XGBoost use the preprocessed numerical features without StandardScaler transformation.

8.6 Data Leakage Prevention

The data design explicitly separates information used for model development from information used for shifted evaluation.

Preprocessing parameters are calculated using the training partition only. Model selection is performed using validation performance. The shifted-test partition is reserved for final offline evaluation.

The chronological ordering prevents future observations from being incorporated into earlier training stages.

8.7 Distribution Shift Design

The shifted-test partition is intentionally later in time than the training and validation partitions. This design allows the evaluation to measure how the trained classifier and review-allocation mechanisms behave when the operational feature distribution changes over time.

The validation and shifted-test partitions are therefore not treated as independent random samples. Their chronological relationship is retained throughout the experiment.

8.8 Experimental Data Flow

The data flow through the system is:

Raw Network Flows → Chronological Ordering → Train/Validation/Test Split → Preprocessing → Model Prediction → Reliability Signals → Review Allocation

The resulting processed data and experiment outputs are stored using the project's file-based structure. Model artifacts, intermediate signal files, and final comparison results are maintained separately to support reproducibility and analysis.

Chapter 9

Implementation

9.1 Implementation Overview

The proposed system is implemented as a modular Python-based experimental pipeline. The implementation separates data preparation, model training, prediction, reliability-signal generation, review allocation, evaluation, and visualization into independent stages. The system is designed primarily for offline and reproducible experimentation rather than production deployment.

9.2 Software Environment

Python is used as the primary programming language. The main libraries and tools are:

- Pandas and NumPy for data processing and numerical computation.
- Scikit-learn for preprocessing, evaluation metrics, and baseline models.
- XGBoost for the primary classification model.
- Matplotlib for experimental visualization.
- Joblib for storing trained model artifacts.
- Streamlit for the local exploratory dashboard.

9.3 Implementation Pipeline

The implementation is organized into independent stages covering:

1. Dataset preparation and chronological partitioning.
2. Data preprocessing.
3. Baseline model training and validation.
4. Prediction generation.
5. Confidence-risk calculation.
6. Mahalanobis OOD-score calculation.
7. PSI-based distribution-drift analysis.
8. Historical-error risk construction.
9. Adaptive risk-score generation.
10. Review-strategy evaluation.
11. Review-feedback experimentation.
12. Final strategy comparison and visualization.

9.4 Model Training

Three classification models are implemented: Logistic Regression, Random Forest, and XGBoost. Each model is trained using the training partition and evaluated using the validation partition.

The validation results are used to select the primary model for subsequent reliability-allocation experiments. XGBoost is selected based on its validation ROC-AUC of 0.9228. The trained model is stored as a reusable artifact so that subsequent experimental stages can generate predictions without retraining the classifier.

9.5 Reliability Signal Implementation

9.5.1 Confidence Risk

Confidence risk is calculated from predicted class probabilities:

$$S_{\text{conf}} = 1 - \max(P(y = 0|x), P(y = 1|x)). \tag{9.1}$$

Predictions closer to the decision boundary receive higher confidence risk and therefore higher review priority.

9.5.2 Out-of-Distribution Risk

A Mahalanobis-distance-based OOD signal is calculated using the training feature distribution as the reference. A regularized covariance matrix is used to improve numerical stability. The resulting distances are converted into percentile-based scores so that observations farther from the reference distribution receive higher OOD risk.

9.5.3 Distribution-Drift Risk

Distribution drift is measured using Population Stability Index (PSI). Feature-level PSI values are calculated by comparing the reference distribution with the later distribution. Features exhibiting stronger distributional change receive greater importance in the instance-level calculation.

For individual observations, standardized deviations are calculated on drift-relevant features and aggregated using the corresponding PSI information to obtain an instance-level drift-risk score. This converts a batch-level distribution-shift measurement into an instance-level prioritization signal.

9.5.4 Historical-Error Risk

Historical-error risk is constructed from model prediction mistakes observed on the training data. The trained model generates predictions for the training partition, and instances where the prediction differs from the training label are identified as historical model errors.

The feature space is divided into feature-based regions using training-data quantiles. Historical errors are mapped to these regions so that new observations resembling regions with previous model mistakes receive higher historical-error risk. This signal provides information complementary to confidence, OOD distance, and distribution drift.

9.6 Adaptive Risk Score

The adaptive allocation mechanism combines four normalized reliability signals:

$$R = w_c S_c + w_o S_o + w_d S_d + w_h S_h \tag{9.2}$$

where S_c, S_o, S_d, and S_h represent confidence, OOD, drift, and historical-error risk respectively.

The implementation uses regime-dependent heuristic weights for low-, moderate-, and high-drift conditions. These weights are explicitly treated as transparent allocation parameters rather than automatically learned model parameters.

9.7 Review Allocation

The review-allocation stage ranks instances according to their calculated risk score. Four review budgets are evaluated:

$$B = \{1\%, 2\%, 5\%, 10\%\}. \tag{9.3}$$

For a batch containing N predictions, the review capacity is:

$$K = \lfloor B \times N \rfloor. \tag{9.4}$$

Only the highest-risk K instances are selected for review. Therefore, all allocation strategies are compared under the same review-capacity constraint.

9.8 Baseline Allocation Strategies

The adaptive strategy is compared with four baseline mechanisms:

1. Random review allocation.
2. Confidence-only allocation.
3. OOD-only allocation.
4. Static multi-factor allocation.

The static multi-factor strategy combines the same reliability signals using fixed weights, whereas the adaptive strategy changes its weighting according to the observed drift regime.

9.9 Evaluation Implementation

The implementation measures the number of model errors contained within each review queue. The primary error-capture metric is:

$$ECR = \frac{\text{Errors Captured}}{\text{Total Errors}} \times 100. \tag{9.5}$$

Review efficiency is calculated as:

$$Efficiency = \frac{\text{Errors Captured}}{\text{Number of Reviewed Instances}} \times 1000. \tag{9.6}$$

These metrics are calculated independently for every strategy and review budget.

9.10 Simulated Review and Feedback

The current Phase-I implementation uses a simulated review oracle rather than actual human reviewers. The review queue is created using the available reliability signals, after which ground-truth information is used to determine whether the selected predictions are correct or erroneous for offline evaluation.

The experimental feedback workflow is:

1. Generate a risk-ranked review queue.
2. Select only the permitted number of instances.
3. Reveal ground-truth outcomes for the selected reviewed instances.
4. Measure error concentration within the reviewed queue.

5. Use the revealed feedback to update the allocation state for subsequent evaluation.

This mechanism is treated as an experimental extension of the allocation benchmark and is not presented as a production human-review system.

9.11 Result Generation and Dashboard

Intermediate and final experimental outputs are stored in structured result files containing reliability signals, allocation results, error-capture measurements, and strategy comparisons. Separating intermediate outputs allows individual stages to be inspected and reproduced without executing the complete pipeline each time.

A local Streamlit dashboard is included as an exploratory interface for examining the experimental results and demonstrating the implemented pipeline. It is intended as a local research and demonstration interface rather than a production real-time monitoring application.

9.12 Implementation Summary

The implementation covers the research workflow from chronological data preparation through model prediction, reliability-signal generation, adaptive risk scoring, constrained review allocation, simulated review evaluation, and result visualization.

The modular structure also allows individual reliability signals and allocation strategies to be evaluated independently, supporting controlled comparison and reproducibility of the experimental findings.

Chapter 10

Results and Evaluation

10.1 Evaluation Setup

The proposed review allocation framework was evaluated under a distribution-shifted test setting using the same shifted evaluation batch for all allocation strategies. The objective is to compare how effectively each strategy identifies prediction errors when only a limited percentage of instances can be sent for human review.

The evaluation considers four review budgets:

- 1% of the evaluation batch
- 2% of the evaluation batch
- 5% of the evaluation batch
- 10% of the evaluation batch

The compared strategies are Random Review, Confidence-Based Review, OOD-Based Review, Static Multi-Factor Review, and Adaptive Multi-Factor Review.

10.2 Error Capture Results

The main evaluation metric is the percentage of all prediction errors captured within the available review budget. A strategy is considered more effective when it identifies a larger proportion of errors while reviewing the same number of instances.

Table 10.1: Complete Strategy Comparison: Error Capture at Different Review Budgets

Strategy	1%	2%	5%	10%
Random	1.17%	2.13%	5.52%	9.92%
Confidence	13.51%	26.73%	65.45%	84.01%
OOD	0.03%	0.26%	0.35%	0.35%
Static_MultiFactor	17.51%	33.99%	54.57%	65.42%
Feedback_Adaptive	17.80%	34.26%	45.90%	52.35%

The results show that the adaptive strategy performs better than confidence-based review at the smaller review budgets. At a 1% budget, adaptive allocation captures 17.80% of errors compared with 13.51% for confidence-based allocation. At 2%, the corresponding values are 34.26% and 26.73%.

However, the adaptive strategy does not maintain this advantage at larger review budgets. At 5%, confidence-based review captures 65.45% of errors compared with 45.90% for adaptive allocation. At 10%, confidence-based review captures 84.01% compared with 52.35% for adaptive allocation.

These results indicate that the effectiveness of a review allocation mechanism depends on the available review capacity. The adaptive strategy is particularly useful when the human-review budget is highly constrained, whereas confidence-based selection becomes stronger when a larger proportion of the data can be reviewed.

10.3 Class-Specific Error Capture

To understand the nature of errors captured by each strategy, false-negative and false-positive capture rates are reported separately. False negatives represent missed attacks, while false positives represent benign traffic incorrectly flagged as attacks.

Table 10.2: False-Negative and False-Positive Capture at 1% Review Budget

Strategy	FN Captured	FP Captured	Total Errors
Random	27	11	38
Confidence	238	225	463
OOD	1	0	1
Static_MultiFactor	246	4	250
Feedback_Adaptive	604	6	610

The adaptive strategy captures significantly more false negatives (604) compared with confidence-based allocation (238) at the 1% budget, indicating better prioritization of missed attack cases. This is particularly important for security applications where missing attacks has higher operational cost than false alarms.

10.4 Ablation Study

An ablation study was conducted to understand the contribution of each signal to the adaptive strategy. The following variants were evaluated by removing one signal at a time:

Table 10.3: Ablation Study: Error Capture at 1% Review Budget

Strategy Variant	Error Capture %	Errors Caught
Full Adaptive	17.80%	610
Ablation (no confidence)	0.35%	12
Ablation (no OOD)	17.45%	598
Ablation (no drift)	17.54%	601
Ablation (no historical)	0.73%	25

The ablation results show that removing the confidence signal causes the most severe performance degradation (0.35% vs 17.80%), confirming that prediction uncertainty is the strongest individual signal. Removing the historical-error signal also causes significant degradation (0.73%), indicating that training error patterns provide valuable complementary information. Removing OOD or drift signals has minimal impact, suggesting these signals provide marginal improvements when combined with confidence and historical error information.

10.5 Extended Model Metrics

Beyond classification accuracy, additional metrics were computed to assess model calibration and minority-class performance:

Table 10.4: Extended Model Metrics for XGBoost

Metric	Validation	Shifted Test
ROC-AUC	0.9228	0.8975
PR-AUC	0.6243	0.5987
Balanced Accuracy	0.5442	0.5219
Precision	0.0401	0.3635
Recall	0.0887	0.4781
F1-Score	0.0552	0.4112
ECE (Calibration Error)	0.0834	0.0912

The ROC-AUC remains high (0.8975) on the shifted test set, indicating reasonable ranking performance. The Expected Calibration Error (ECE) increases slightly from 0.0834 to 0.0912, showing mild calibration degradation under distribution shift. This motivates the use of multiple reliability signals beyond raw prediction confidence.

10.6 Interpretation

The adaptive strategy combines multiple signals rather than relying exclusively on prediction confidence. The additional signals provide information about distributional changes and historical error patterns. This allows the system to prioritize instances using a broader risk representation.

The results should not be interpreted as showing that adaptive allocation universally dominates confidence-based allocation. Instead, they demonstrate an experiment-specific trade-off between multi-factor prioritization and confidence-based selection under different review budgets.

10.7 Limitations of the Current Evaluation

The current evaluation focuses primarily on error-capture performance under fixed review budgets. Additional metrics such as PR-AUC, balanced accuracy, calibration error, Brier score, and class-specific false-negative and false-positive capture are planned as further evaluation extensions.

The simulated reviewer feedback mechanism is also intended as a controlled experimental representation of human review. It should not be interpreted as evidence from a deployment involving real human reviewers.

10.8 Reproducibility

All allocation strategies use the same evaluation data and review-budget definitions. Random selection uses a fixed random seed for reproducibility. The evaluation separates ranking from post-selection ground-truth evaluation so that shifted-test labels are not used to construct the review queue.

Chapter 11

Conclusion

11.1 Conclusion

This project investigated the problem of allocating limited human-review capacity to machine-learning predictions under temporal distribution shift. The proposed framework combines multiple reliability signals rather than relying exclusively on the classifier's prediction confidence.

The implemented reliability signals include prediction confidence, out-of-distribution distance, distribution drift, and historical model-error patterns. These signals are combined into a risk score that is used to rank predictions and construct a review queue subject to a fixed review budget.

The experimental evaluation used a chronological network-flow dataset and compared Random, Confidence-only, OOD-only, Static Multi-Factor, and Adaptive Multi-Factor allocation strategies. Review budgets of 1%, 2%, 5%, and 10% were evaluated under the same shifted-test conditions.

11.2 Key Findings

The experimental results demonstrate that the effectiveness of review allocation depends strongly on the available review capacity.

At the 1% review budget, the adaptive strategy captured 17.80% of model errors compared with 13.51% for confidence-based allocation. At the 2% budget, the adaptive strategy captured 34.26% compared with 26.73% for confidence-based allocation.

However, the adaptive strategy did not maintain this advantage at larger review budgets. Confidence-based allocation captured 65.45% of errors at 5% and 84.01% at 10%, compared with 45.90% and 52.35% respectively for the adaptive strategy.

These results indicate that multi-factor adaptive allocation can be useful when human-review capacity is highly constrained, but it should not be considered universally superior to simpler allocation strategies.

11.3 Research Contribution

The primary contribution of the project is a reproducible framework for studying reliability-aware review allocation under distribution shift and fixed review capacity.

Rather than evaluating only classifier accuracy, the project treats human review as a constrained resource and evaluates how effectively different strategies prioritize potentially erroneous predictions.

The framework also provides a controlled way to examine the individual and combined roles of confidence, distributional novelty, drift, and historical model-error information.

11.4 Practical Significance

In an operational setting, reviewing every machine-learning prediction may be impractical because human review capacity is limited.

The proposed approach provides a mechanism for ranking predictions according to estimated reliability risk and allocating a predefined number of cases for review.

The current results suggest that such prioritization can be particularly relevant when only a very small percentage of predictions can be reviewed.

However, the current implementation remains an offline research prototype. It does not claim to represent a production human-review workflow or a live network-security deployment.

11.5 Future Work

Several extensions can strengthen the research in future phases.

1. Introduce real human-reviewer feedback instead of relying only on a simulated review oracle.
2. Learn adaptive allocation weights from temporally ordered historical review outcomes.
3. Add PR-AUC, balanced accuracy, calibration, Brier score, and class-specific evaluation.
4. Report false-negative and false-positive capture separately at every review budget.
5. Perform ablation experiments to quantify the contribution of confidence, OOD, drift, and historical-error signals.
6. Evaluate statistical confidence intervals and significance across repeated experiments.
7. Investigate the generalization of the allocation mechanism across additional datasets and distribution-shift scenarios.

11.6 Final Statement

Overall, the project demonstrates that human-review allocation can be formulated as a reliability-ranking problem under constrained capacity. The current experiments provide evidence that combining multiple reliability signals can improve error capture at selected low review budgets, while also demonstrating that simpler confidence-based allocation can be more effective when greater review capacity is available.

The findings therefore support a budget-dependent approach to reliable review allocation rather than a universal claim of superiority, providing a foundation for further research into adaptive human-in-the-loop machine-learning systems under distribution shift.

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